

Hao Zheng

Recent M.S. Graduate, Tsinghua University | Associate Member, PIARC Technical Committee 1.1

✉ scholarhaozheng@gmail.com | zhengh22@mails.tsinghua.edu.cn | github.com/scholarhaozheng | scholarhaozheng.github.io

PUBLICATIONS

- **Hao Zheng**, Han Zheng, George Giannopoulos. KNOWCAST: Domain-knowledge-integrated deep learning for metro OD forecasting and path flow estimation. DOI:10.1016/j.trc.2026.105710 [🔗](#) *Transportation Research Part C: Emerging Technologies*, 189, 105710. 2026
 - **Paper's Contribution:** Proposed a domain-knowledge-integrated deep learning framework for jointly forecasting short-term metro OD flows and inferring path flows, with path-flow estimation structurally coupled to OD forecasts through embedded transportation-domain knowledge.
 - **My Contribution:** Conceptualized the framework, implemented the deep learning model, and led experimental validation, revision, and manuscript preparation.
- **Hao Zheng**, Han Zheng*, George Giannopoulos. Integrated Routing of Trucks and Robots for Multi-Modal Urban Express Logistics. [🔗](#) (Presented at the TRB 105th Annual Meeting, 2026; R1 under review at Transportation Research Record) 01/2026
 - **Paper's Contribution:** Proposed LT-ADR, a logistics system for multi-modal urban delivery that jointly optimizes truck and robot routes for both regular and instant express services. The system is modeled on a multilayer space-time network and solved by a Branch-and-Price algorithm, demonstrating substantial improvements in efficiency and computational speed over commercial solvers.
 - **My Contribution:** Formulated the multi-modal routing problem on a multilayer space-time network and developed the Branch-and-Price algorithm to solve it efficiently.
- Jiantao Li, Yian Zhao, **Hao Zheng**, Runhua Guo. Vehicle Vibration Modeling via Road Profile Reconstruction and Simulation: Coupled Analysis of IRI, Speed, and Ride Comfort for Driving Assistance. *Road Materials and Pavement Design*. 2025
- **Hao Zheng**, Xingchen Zhang, Junhua Chen. Study on Customized Shuttle Transit Mode Responding to Spatiotemporal Inhomogeneous Demand in Super-Peak. *Information*, 2021, 12:429. DOI:10.3390/info12100429. [🔗](#) 2021
 - **Paper's Contribution:** Proposed a customized shuttle bus transportation model to address instantaneous mega-traffic flows with uneven demand. The problem is formulated as a Pickup and Delivery Problem with Time Windows and Transfers (PDPTWT) and solved by a heuristic algorithm combining Tabu Search and ALNS. The method demonstrated significant computational speedup over commercial solvers and achieved substantial reductions in fleet cost, vehicle waiting time, and passenger waiting time in reality-based case studies.
 - **My Contribution:** Formulated the PDPTWT model and developed the hybrid ALNS-TS heuristic algorithm to solve it.
- **Hao Zheng**, Yun Wang*, Xuedong Yan, Yunwei Li. Dynamic Planning Model and Solution Algorithm for the Demand-responsive Customized Bus Route. *World Transport Convention (WTC) Proceedings*, 2021. [🔗](#) 2021
 - **Paper's Contribution:** Developed a two-stage planning model and solution algorithm for customized bus services, enabling dynamic route adjustments to accommodate real-time passenger demands on top of existing static reservations.
 - **My Contribution:** Modified the core algorithm, implemented the solution in code, conducted computational case studies for validation, and was the primary author of the manuscript.
- Yuyan Tan, Yibo Li, Ruxin Wang, Xiwei Mi, Yaxuan Li, **Hao Zheng**, Yu Ke, Yan Wang. Improving Synchronization in High-Speed Railway and Air Intermodality: Integrated Train Timetable Rescheduling and Passenger Flow Forecasting. *IEEE Transactions on Intelligent Transportation Systems*, 1–17. DOI:10.1109/TITS.2021.3137410. 2022

Book

- Huapu Lu, Fang Wang, et al., and **Hao Zheng**. *Green Development Paths and Strategies for Comprehensive Transportation in Ningxia*. 2024

MANUSCRIPTS UNDER REVIEW

- **Hao Zheng**, Yian Zhao, Jiantao Li, Xi Zhu, Runhua Guo*. An Operational Scheduling Framework for Pavement Maintenance Activities: From Rule-Based Task Generation to Branch-and-Price-Based Route Optimization. [🔗](#) (Submitted to International Journal of Pavement Engineering) 04/2026
 - **Paper's Contribution:** Developed an integrated operational scheduling framework for pavement maintenance activities, linking rule-based task generation with branch-and-price-based route optimization. The method reconstructs maintenance tasks on road segments into an optimization-friendly topology and formulates the execution stage as a Vehicle Routing Problem with Time Windows (VRPTW), solved using a custom Branch-and-Price algorithm.
 - **My Contribution:** Led model development, coding, experimental validation, and manuscript writing.

- **Hao Zheng**, Han Zheng*, George A. Giannopoulos. DISCERN: Discrete Choice with Embedded Representation Networks. *(In preparation, 2026; target journal: Transportation Research Part C)* 2026
 - **Paper's Contribution:** Proposed DISCERN, a framework that calibrates the behavioral parameters of traditional Discrete Choice Models (DCMs) using large-scale macroscopic data. This is achieved by embedding a theory-driven DCM within a deep learning architecture and using backpropagation on a macro-level predictive task to infer latent micro-level parameters.
 - **My Contribution:** Led model development, coding, experimental validation, and manuscript writing.
- **Hao Zheng**, Runhua Guo*. STAGED TRANSFER LEARNING FOR LOCAL PAVEMENT CONDITION PREDICTION UNDER HETEROGENEOUS INDICATORS.  *(In preparation, 2026; planned submission to Journal of Transportation Engineering, Part B: Pavements) 2026*
 - **Paper's Contribution:** Proposed a staged transfer learning framework for local pavement condition prediction under heterogeneous indicators. The method leverages pre-training on larger external data and progressive adaptation to local data, improving reliability under data scarcity and indicator mismatch.
 - **My Contribution:** Designed and implemented the staged transfer learning framework and conducted model development, experimental validation, and manuscript writing.
- **Hao Zheng**, Yuhan Tang, Nina Cao, Yibo Zhao, Xuan Jiang*, Dingyi Zhuang, Shenhao Wang, Haris N. Koutsopoulos, and Jinhua Zhao. A Bilevel Optimization Framework for Collaborative Strategies between Infrastructure and Mobility Service Providers.  *(In preparation, 2026; target journal: Transportation Research Part C)* 2026
 - **Paper's Contribution:** Developed a bi-level optimization framework integrated with a high-performance, GPU-based microscopic traffic simulator (LPSim) to model and solve strategic interactions (e.g., tolling and routing) between infrastructure providers and AV fleet operators. Analyzed multiple collaboration scenarios to quantify trade-offs between operator revenue and system-wide efficiency.
 - **My Contribution:** Designed and implemented the end-to-end bi-level optimization framework; developed the Python-based upper-level optimization loop and re-engineered the C++/CUDA-based lower-level simulator to integrate toll-responsive agent logic.

RESEARCH EXPERIENCE

- **Computational Benchmarking of Scalable Traffic Assignment on Open Transportation Networks** 2025 – Present
Research Collaborator with Prof. Xuesong Zhou
 - Built an experimental pipeline on open transportation network benchmarks by converting TNTP-style instances into unified link-demand formats and generating path sets for path-based traffic assignment.
 - Validated a path-compression framework for large-scale traffic assignment under different compression ratios and demand levels, comparing its computational efficiency and solution fidelity against baseline solvers.
 - Analyzed speed-accuracy trade-offs using variable reduction, pure solver time, objective gap, and link-flow fit metrics on open network instances.
- **A Bilevel Optimization Framework for Tolling Strategy in Autonomous Networks**  06/2025 – 10/2025
MIT-UF-NEU Summer Research Camp / Supervisor: Prof. Jinhua Zhao, Mentor: Dr. Xuan Jiang
 - Project: Understanding different cooperation patterns between AV companies and road infrastructure companies.
 - Designed and implemented an end-to-end bi-level optimization framework to find optimal tolling strategies by modeling the interaction between infrastructure operators and autonomous vehicle fleets.
 - Developed the upper-level model in Python to create an automated optimization loop that configures and executes a traffic simulator, processes results, and iteratively updates tolling policies based on revenue and congestion feedback.
 - Re-engineered a high-performance C++/CUDA traffic simulator to serve as the lower-level model, implementing toll-responsive logic and a cost-benefit analysis for agent lane-changing behavior.
 - Verified the framework's effectiveness while generating substantial revenue across various tested scenarios.
- **Digital Platform for Intelligent Road Asset Management (Code Writing: Prediction & Decision Models)** 2023 – 2025
Project: Research on Integration and Application of Intelligent Air-Ground Equipment in Digital Transportation Management
 - Developed the core algorithms for a comprehensive digital road asset management platform built on a SpringBoot + VUE framework.
 - Designed and implemented the road performance prediction module, creating a deep learning model using GRU and attention mechanisms that incorporates climate, traffic, and material data, and trained the model on the LTPP dataset.
 - Built the maintenance optimization module, which uses a mathematical model under dual budget and performance constraints to generate optimal, fine-grained maintenance strategies.
- **Strategic Framework for Green Transportation & Infrastructure Management (Researcher & Contributing Author)** 2022 – 2024
Project: Coordinating Ecological Protection with Carbon Peaking and Neutrality Goals in the Yellow River Basin / Supervisor: Prof. Runhua Guo

- o Analyzed infrastructure lifecycle management strategies, including intelligent inspection methods, pavement performance prediction, and maintenance decision-making models.
 - o Under the primary guidance of Prof. Guo, authored two key chapters (Chapter 3: Green Construction; Chapter 4: Green Operations & Maintenance) of the project’s resulting book, published by China Communications Press.
- Comparative Analysis of Road Design Standards for Belt and Road Projects (Research Analyst)** 2023
Project: Differentiated Research on Engineering Projects along the Belt and Road: Sri Lanka / Supervisor: Prof. Runhua Guo
 - o Conducted a comparative analysis of standards for highway construction and acceptance between China and Sri Lanka to provide technical guidance for international projects.
 - o Authored a technical report identifying critical differences in design parameters (e.g., standard axle loads, material specifications, and reliability levels) to facilitate the adaptation of Chinese engineering practices abroad.
- Optimizing Air-Rail Intermodal Transport: A Case Study of Daxing Airport (Project Leader)** 2020 – 2022
National University Student Innovation & Entrepreneurship Development Program / Supervisor: Prof. Yuyan Tan
 - o Led a student team to design a two-stage optimization framework aimed at improving the end-to-end efficiency of air-rail passenger transfers.
 - o Formulated a multi-objective optimization model to adjust high-speed rail timetables, maximizing flight connection opportunities while minimizing schedule disruptions.
 - o Designed a customized bus routing solution for the airport-station connection, modeling it as a VRPTWTS and developing a hybrid Adaptive Large Neighborhood Search and Tabu Search (ALNS-TS) algorithm to solve it.
 - o Implemented the solution as the Multi-Type Customized Shuttle Bus Route Planning Software, for which I was granted a national software copyright as the sole holder (Reg. No. 2021SR1289113).
 - o The project received a national-level award, and the research was accepted for publication in the World Transport Convention (WTC) proceedings.

PROFESSIONAL SERVICE

- Associate Member**, PIARC Technical Committee 1.1 (World Road Association) 04/2026 – Present
- Contributor**, PIARC TC 1.1 Working Group 1 Survey Report: *Envisioning the Transport Agency of the Future* 04/2026 – Present

EDUCATION

- Tsinghua University**, School of Civil Engineering, Master of Science, GPA: 3.86 09/2022 – 06/2025
- Beijing Jiaotong University**, School of Traffic and Transportation, Bachelor of Engineering, GPA: 3.80 09/2018 – 06/2022

THESES

- Master Thesis:** A Comprehensive Full-Process Planning Model for Pavement Performance Forecasting and Maintenance Strategy Implementation under Missing Data Conditions. 06/2025
- Undergraduate Thesis:** Customized Bus Planning Method Responding to Spatiotemporal Concentrated Demand in Super-peak. 06/2022

SKILLS

- Programming Skills:** Proficient in Python, MATLAB, and C#; experienced with C++/CUDA and GAMS.
- English Proficiency:** TOEFL: 112 (Reading: 30, Listening: 30, Speaking: 25, Writing: 27) 01/2025

AWARDS & HONORS

- Second-Class Academic Excellence Scholarship, Tsinghua University 2024
- National Level Award, National University Student Innovation & Entrepreneurship Program 2022
- Outstanding Graduate of Beijing Municipality (Top 5%, 19/380) 06/2022
- National Software Copyright (Sole Holder), Reg. No. 2021SR1289113 2021
- “Outstanding Student in Transportation”, China Institute of Communications Education 11/2021
- Wu Fu-Zenhua Transportation Education Excellent Student Award (Top 0.1%, 1/1520) 12/2021
- Zhan Tianyou Scholarship (Top 0.2%, 2/1520) (Awarded Twice) 12/2021, 12/2020
- Zhixing Scholarship (President’s Scholarship) (Top 0.1%, 10/16800) 12/2020

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| • University-Level Merit Student (Awarded Twice) | 12/2021, 12/2020 |
| • Excellent Academic Scholarship (Awarded Three Times) | 12/2021, 12/2020, 12/2019 |